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Warm glow of giving collectively – An experimental study

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ABSTRACT

We report on an experiment to test for a warm glow of giving collectively. Comparing subjects' affective state before and after the experiment, we find that individual charitable donations create a feeling of warm glow while collective donations do not. Proposing to donate the full endowment collectively improves subjects' affective state significantly, though the behavioral data suggests that this result cannot be explained by the notion of expressive voting. We also find that subjects who consider Kant's Categorical Imperative to be an important guideline for individual decisions are more likely to donate the full endowment to charity. This result supports the notion of Kantian thinking as an independent factor explaining cooperative behavior (Roemer, 2014).

1. Introduction

Altruistic motives are recognized to play an important role in motivating pro-social behavior. The concept of pure altruism assumes that individuals are motivated by the wish to help the needy (e.g. Roberts, 1984; Warr, 1982). Here, prosocial behavior is motivated by empathy, i.e. the "genuine concern for the situation of someone else" and the intention to improve the well-being of the needy (Schokkaert, 2006: 134). Andreoni (1988) argues that people may experience a good feeling "like a warm glow" from the very *act* of giving (e.g. Andreoni, 1988). This feeling provides an independent motivation for pro-social behavior that does not refer to the utility of the supported but solely to the immediate good feeling the supporter receives from the very act giving support. A number of empirical studies provide evidence for an idiosyncratic warm glow effect (e.g., Crumpler & Grossman, 2008; Korenok, Millner, & Razzolini, 2013). Konow (2010) runs an experiment based on variations of the dictator game. He uses a psychological scale to measure participants' short-run affective state at the beginning of the experiment and after they have made their decisions to share resources in the dictator game. The basic hypothesis behind this approach is the following: If giving to others creates a feeling of warm glow, there must be a positive relationship between the amount given in the dictator game and the change in short-run affective state. His observations support this hypothesis: Giving has a positive impact on the donor's short-run affect if giving means a donation to an organization that helps children in need.

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In many countries, a substantial amount of transfers to the poor is organized by the state and decided about in democratic procedures. Thus, it is interesting to ask whether there is a warm glow of giving collectively. Essentially, this question matters for the same reason that the question about the warm glow from individual contributions to public goods matter: Within a country, a possible warm glow influences the way in which municipal budgets react to earmarked conditional grants for pro-social services. If there is no warm glow from collective giving, earmarked pro-social funds from federal or regional governments crowd out the funds the municipality itself is willing to reserve for these purposes. If, however, there is a warm glow from collective giving, crowding out is incomplete.[2] In developing democratic countries, a similar argument applies to private funds from outside the country, e.g. from foundations supporting pro-social activities or from supranational institutions like the United Nations. Thus, knowing whether or not there is a warm glow of collective giving helps to better predict government decisions and thus design better policies. So far, however, a possible warm glow from collective giving has received little attention. Harbaugh, Mayr, and Burghart (2007) analyze the neural activity when individuals are forced to donate collectively. The subject in one treatment are taxed to finance donations to charity while the subject in the other treatment decide voluntarily and on an individual basis to donate an equivalent amount of money to charity or to keep it for private use. In their study, subjects classified as altruists show reward-related neural activity even when being forced to give the money to charity. The activation is higher when charitable giving is voluntary than when transfers are mandatory. Nevertheless, this result suggests that there may be a warm glow giving even when subjects did not have any influence on the amount donated – not even as a group.

In this paper, we want to test whether there is a warm glow from giving collectively when the decision about how much to donate per capita is reached collectively. We apply the methodology proposed by Konow (2010): We run an experiment that compares the change in short-run affect from individual and collective charitable donations in a unified between-subject setting. Every subject is given an initial endowment of 10 €. In treatment COL, the group decides collectively how much of this endowment every group-member must donate to charity. Every subject proposes an amount for the per-capita donation. In the end, every participant has to donate the median amount proposed. In treatment IND, every individual can decide for himself how much of his endowment to donate.

Like Konow (2010), we find that donating to charity improves individuals' short-run affective state. The higher the donation, the larger is the improvement in affective state. This result suggests that there is a warm glow from giving the full 10 €. In treatment COL, subjects' change in affective state is not found to depend on the size of the collective donation they expect to make in the end. Thus, we find no evidence for a warm glow from collective giving. On the other hand, we find that proposing to donate the full 10 € to charity improves subjects affective state significantly while there is no relationship between the amount proposed and affective state for amounts below 10 €. By itself, this result seems to support the theory of expressive voting (e.g., Hamlin & Brennan, 1998; Tullock, 1971). However, the behavioral pattern of donation is not line with this theory. Summing up, proposing to donate the full 10 € is found to be an individually rewarding act while we have little to say about the factors that motivate some people to propose the full 10 € for donation and others do propose less. As a side-result, we find that subjects who consider Kant's Categorical Imperative to be an important guideline for individual decisions are more likely to donate the full endowment to charity. This result supports the notion of Kantian thinking as an independent factor explaining cooperative behavior (Roemer, 2014).

The paper proceeds as follows: Section 2 describes the experimental set-up. Section 3 presents the central hypothesis. Results are described in Section 4. Section 5 concludes.

2. Experimental set-up

The experiment involves two treatments IND and COL. Every subject makes one decision during the experiment (see Appendix C for instructions). The decision task can be described as follows: At the beginning of the experiment, every subject receives a voucher worth 10 €. Under treatment IND, subjects are asked to decide how much of the initial endowment to donate to the Kinderschutzbund (KSB), a large German NGO that uses funds to help children in need or threatened by violence. Every subject states the amount X he wants to donate. This amount is donated on his behalf and he can cash in the remaining $10 - X$ €. Under treatment COL, subjects have to make a collective decision about how much of the individual endowment every subject in the group has to donate to the KSB. Every participant is asked to propose an amount X to be donated per capita. In the end, every participant in treatment COL has to donate the median amount X^M proposed in his group. The remaining $10 - X^M$ € can be cashed in.

The experimental procedures are identical for both treatments: All subjects involved in one session take the decision simultaneously. They sit in one classroom so that can see their fellow-players. Instructions are given in written form and communication is prohibited throughout the experiment. Decision sheet and questionnaires are handed out on paper and subjects fill them in during the session in the classroom. We answer arising questions on the instructions in private with the individual. A session lasts about 30 min. We guarantee anonymity during the experiment. To minimize social pressure or experimenter demand effects (Bischoff & Frank, 2011; Zizzo, 2010), we use sealed envelopes to pay subjects. The envelopes have been filled days before the subjects receive them.

During the experiment, the participants are asked to fill in two questionnaires – one at the beginning of the session and one at the end of it. We use the short version of the Positive Affectivity Negative Affectivity Schedule (PANAS) proposed by

[2] Thereby, warm glow from collective giving may provide a new explanation for the flypaper effect (Inman, 2008).

MacKinnon et al. (1999) to measure the current affective state of subjects at the beginning of the experiment as well as immediately after they have made their decision to donate respectively stated their proposal for a collective donation. In the second questionnaire, we ask subjects for biographical information and their expectations with respect to the behavior of fellow-participants. The estimates for the average donation (treatment IND) resp. the median amount proposed for the collective donations (treatment COL) are incentivized. In each session, the subject with the most accurate prediction receives an extra 10 €. Participants who answer all questions in the questionnaires are paid an extra amount of 1 €.

The sessions took place in November 2011 at the University of Kassel, Germany and – a few days later – at the University of Applied Sciences Anhalt (Bernburg), Germany. Subjects were recruited in a lecture on "math for economists" in Kassel and in an introductory lecture of economics in Bernburg. Both lectures were held by lecturers who are not involved in the experimental project. Students were informed that participation is voluntary but only very few of them decided not to participate. One advantage of conducting the experiment in classroom is that it is expected to reduce the sample selection bias in comparison to other laboratory experiments where participants are first recruited for an experiment conducted at a later date. In each session, we assigned subjects at random to one of the treatments and ran the corresponding sessions simultaneously.

3. Hypotheses

Inspired by Konow (2010), we focus on the impact of subjects' decisions and expectations on their short-run affective state. Short-run affect captures the way that an individual feels at the moment. Affect is the supraordinate concept that includes mood and emotions (e.g. Kleinginna & Kleinginna, 1981; Marcus, 2000). An emotion is psychophysiological experience that is caused by an external impulse or sensation. Emotions can be "attributed do a concrete sensation or experience" (Steenbergen, 2010) while mood is defined as a "general and pervasive feeling state" that exists without explicit reference to its sources (Marcus, 2000: 224). Konow (2010) measures subjects' short-run affective state using a scale proposed by Batson et al. (1988). Specifically, he asks subject to express their current affect in the dimensions *elated – depressed* and *good mood – bad mood* (10 point Likert-scale). He adds the change in score across both dimensions to arrive at his indicator for change in short-term affective state. The PANAS-concept we use provides subjects with a list of five adjectives that describe how one can feel and asks them to state the degree to which they feel this way right now on a 5-point scale (MacKinnon et al., 1999). The answers are aggregated to derive an indicator for subjects' (positive) affective state.[3] The main advantage of these indicators compared to the one used by Konow (2010) is that – by aggregating across a five items – the PANAS indicators provide a more reliable picture of subjects' affective state.

While most scholars would agree that utility and affect are related, the precise relationship between the two concepts is still disputed (e.g., Kahneman, Wakker, & Sarin, 1997; Kimball & Willis, 2006). For the purpose of our study, it is sufficient to assume that a difference between the affective states reported before and after our experiment is causally related to subjects' experiences during the experiment. Warm glow is explicitly defined by positive feelings coming along with giving. Thus, we follow Konow (2010) and hypothesize that subjects witness an improvement in short-run affective state if they donate money. The improvement is expected to increase in the amount donated. Hypothesis H1 thus reads:

H1. In treatment IND, the change in short-run affective state is increasing in the amount X donated.

In treatment IND, subjects have perfect control over the outcome. If they decide to donate X €, X € is donated. In other words, intention and outcome are congruent. This is entirely different for treatment COL. An individual who intends to donate X € and proposes to do so cannot be sure that the collective decision leads to the intended outcome. The difference between intended outcome and real outcome is especially large for individuals whose preferences differ from the preferences of the median participant. Thus, in treatment COL, the warm glow of giving does not depend on the amount X the individual proposes for the collective donation. Instead, it depends on the amount the individual expects to donate in the end. This in turn depends on individual's expectations concerning the median amount proposed for the collective donation. This brings us to Hypothesis H2:

H2. In treatment COL, the change in short-run affective state is increasing in the amount that an individual expects to donate.

Hamlin and Brennan (1998) argue that citizens experience a good feeling when expressing their opinion on political issues (see also Hamlin & Jennings, 2011). It is commonly assumed that this so-called expressive utility is higher for expressing approval to redistribution than for rejecting redistributive policies (e.g. Shayo & Harel, 2012; Tullock, 1971; Tyran, 2004).[4]

[3] The PANAS scale contains another five adjectives to measure negative affectivity. As warm glow is explicitly defined by positive feelings experienced during the experiment, we concentrate on the changes in the scale for positive affective state.

[4] The empirical support for the importance of expressive motives is mixed. While early experimental studies do not find support for expressive motives (e.g., Carter & Guerette, 1992; Fischer, 1996; Tyran, 2004), more recent experiments support importance of expressive motives (e.g., Bischoff & Egbert, 2013; Shayo & Harel, 2012). There is an independent strand of literature in voter behavior that focuses the question whether strategic voting may undermine the majority votes's ability to aggregate private information as proposed by Condorcet's Jury Theorem (e.g., Battaglini, Morton, & Palfrey, 2010). In the corresponding experiments, majority voting is a means of aggregating private information when individuals share a common goal but disagree about the correct way to pursue it. In treatment COL of our experiment, voting serves as a means of aggregating heterogeneous individual preferences for pro-social policies. There is no consensus about the correct policy. More importantly, there is no room for strategic voting in our experiment. Thus, the main questions raised by Battaglini et al. (2010) and other authors in this strand of literature do not apply to our experiment.

According to the theory of expressive voting, subjects can attain the expressive utility from proposing a high collective donation at negligible costs as long as they do not expect to be pivotal (e.g., Tyran, 2004). In our experimental setting, the incentive to propose a high collective donation even holds for individuals who expect to be pivotal. To understand why, consider the situation an individual subject i deciding about his proposal for the collective donation. Ex ante, the probability for a certain individual to be the median proposer is very small (e.g., Beck, 1975). And even if he is pivotal, he does not choose between option "no collective donation" and "donating a certain amount X per capita". Instead, a change in the median proposer's proposal X_i only influences the per capita donation X^M as long as this change takes place between the limits of the next-smallest value (X^{M-1}) and the next-largest value (X^{M+1}) proposed. The largest increase in the collective donation that the median proposer can induce by deviating from his original proposal X_0^M is a plus of $X^{M+1} - X_0^M$ respectively a minus of $X_0^M - X^{M-1}$. Any change beyond these limits means that he loses the position as median proposer and has no further impact. The more subjects participate, the smaller the range $X^{M+1} - X^{M-1}$. In large elections, it is very likely that $X^{M-1} = X_0^M = X^{M+1}$ and thus a change in an individual proposal does not change the collective donation finally made even if this individual is (one of) the median proposer(s). In the very likely case that an individual is not the median proposer, he can only influence X^M if he jumps from proposing a value below X^M to one above X^M (or the other way round). This changes the median proposer and may increase (decrease) X^M. However, even then the change in X^M is much smaller than the change the individual makes in his proposal X. Thus, all subjects whose utility function contain an argument for the expressive utility from proposing a large collective donation can attain this utility at negligible material costs – regardless of whether they expect to be pivotal or not. In essence, the theory of expressive voting implies that the amount X an individual proposes to donate does not reflect the amount he intends to donate collectively, nor does it represent the amount he would want to donate privately if he was in treatment IND. It represents the proposal that maximizes his utility – knowing that his proposal has no impact on the true outcome. In the current experiment, donations support children in need and threatened by violence – a charitable purpose that is extremely well-accepted. Thus, we expect that the expressive utility from proposing a donation for the KSB increases in the amount proposed. This lead to Hypothesis H3:

H3. In treatment COL, the change in short-run affective state is increasing in the amount X proposed, other things being equal.

4. Results

A total of 412 undergraduate students participated. Table 1 reports the average and median donations for both treatments as well as on the expected donation per capita.

Fig. 1 contains histograms for the individual donations resp. the collective donations proposed. The distribution is strikingly similar. In both treatments, about 30% of subjects donate respectively propose the full 10 €. In both treatments, far more than 50% of all participants, donate or propose 5 € or more. This indicates that the purpose of the donation – helping children in need or threatened by violence – strongly triggers motives beyond material self-interest. To some extent, the detailed information on KSB provided before subjects made their decision may have contributed to this (e.g. Milinski, Semmann, Krambeck, & Marotzke, 2006; Zizzo, 2010) We find no significant difference between the average individual donation in treatment IND and the average collective donation proposed in treatment COL.

4.1. Analyzing the change in short-run affective state

Table 1 also reports the average and median change in affect by treatment. The change is positive on average and there are no differences across treatments. We hypothesize that the affective state improves with subjects' donations respectively proposals for the per capita donations stated in the experiment (see Hypotheses H1 and H2). We test these hypotheses in regressions that use the change in positive short-run affective state ΔSRA that subjects witness during the experiment as endogenous variable. Following the standard procedure in the psychological literature, ΔSRA is measured by the difference in average short-run-affectivity score at the beginning of the session and at the end of it (e.g., MacKinnon et al., 1999). Though ΔSRA is truncated by definition, there are no observations at the truncation limits. Therefore OLS regressions are used.

In treatment IND, the individual donation IND_DONATED is the primary explanatory variable (test for warm glow in treatment IND (H1)). To account for the possibility that the change in affective state also depends on the total amount raised in the experiment, we include the average donation IND_AV_DONATION_EXP as expected by subject i. In the regression for treatment COL, we include the expected per capita donation COL_DONATION_EXP (test for warm glow in treatment COL (H2)) as well as the individually proposed per capita donation COL_PROPOSED as explanatory variable (test for expressive motives in treatment COL (H3)). We also account for the fact that – unlike in treatment IND – subjects in treatment COL only have limited control over how much they donate in the end. Some subjects in treatment COL expect to be forced to donate more than they want to, while other subjects expect the collective per capita donation to be lower than the one they proposed. In both cases, this may have a negative impact on these subjects' affective state after the experiment and thus – other things equal – on ΔSRA. We control for this by including the difference between individual proposal and expected donation COL_FRUSTRATION = |COL_PROPOSED − COL_DONATION_EXP| in the regression. In addition, we control for subjects' affective state at the beginning of the experiment (SRA_0).

Table 1
Descriptive statistics.

Treatment	IND	COL	ALL
N	210	202	412
N (Kassel)	152	144	296
N (Bernburg)	58	58	116
Average *X*	5.56	6.01	5.78
Median *X*	5	5	5
Average ΔSRA	.2331	.1468	.1907
Female participant (%)	40%	53%	49%
Average age	21.88	22.12	22

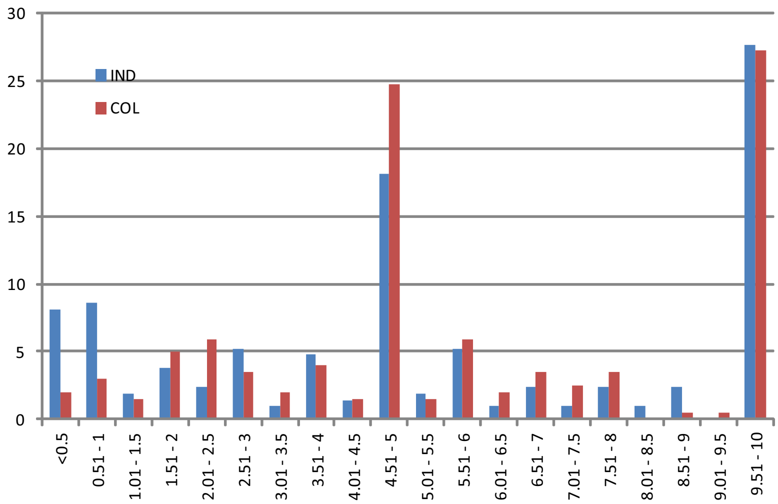

Fig. 1. Histogram of individual donations resp. proposals (in %).

The results are reported in the first two columns of Table 2. In both treatments, the individually donated (IND_DONATED) respectively proposed (COL_PROPOSED) amount has a positive and highly significant impact on ΔSRA. The affective state reported at the beginning of the experiment SRA_0 produces a significantly negative coefficient estimator in both treatments. In treatment COL, the expected median donation COL_DONATION_EXP and COL_FRUSTRATION do not cause a significant change in subjects' short-run affective state.

Given that approximately 30% in both treatments donate respectively propose do donate the full 10 €, it is possible that the observed relationships are primarily driven by those who gave the full 10 €. We test for this possibility by splitting the variables IND_DONATED and COL_PROPOSED into two variables. The dummy variable IND_DONATED = 10 takes on the value 1 for individuals that donate the full 10 € (0 else). The continuous variable IND_DONATED_LESS is equal to the variable IND_DONATED for individuals who donate only part of their endowment but takes on the value 0 for subjects donating the full 10 €. The variable COL_PROPOSED is split in an analogous way. The regression results for the modified models are reported in columns 3 and 4 of Table 2. In both treatments, donating respectively proposing the full 10 € has a significantly positive impact on subjects' short-run affective state. The size of the coefficient estimator for IND_DONATED = 10 resp. COL_PROPOSED = 10 in columns 3 and 4 is largely identical to the coefficient estimator for IND_DONATED resp. COL_PROPOSED multiplied by 10. For donations below 10 € (treatment IND), the impact on the affective state remains significantly positive while proposing less than 10 € (treatment COL) is not found to have an impact on subjects' affective state. This result supports the notion that the performance of COL_PROPOSED is largely driven by those subjects who proposed the full 10 €.

The performance of COL_PROPOSED and COL_PROPOSED = 10 supports the main thrust of Hypothesis H3. This result by itself is in line with the notion of expressive voting. However, the theory of expressive voting has a second implication that is not at all in line with the empirical pattern observed in this experiment: If many subjects' utility function contains an element of expressive utility, the main argument of Gordon Tullock's "Charity of the uncharitable" applies and the average proposal made in treatment COL should exceed the average amount donated in treatment IND (Tullock, 1971). Fig. 1 provides absolutely no support for this notion. The corresponding test show that the average and median values for COL_PROPOSED and IND_DONATED as well as the share of subjects stating the full 10 € are far from being significantly different. Thus, the

Table 2
OLS-regressions to explain the change in short-run positive affectivity.

End. var. ΔSRA	IND	COL	IND	COL
IND_DONATED	.0609(.014)***			
IND_DONATED = 10			.609(.144)***	
IND_DONATED_LESS			.0492(.020)**	
COL_PROPOSED		.0407(.017)**		
COL_PROPOSED = 10				.409(.171)**
COL_PRPOSED_LESS				.0072(.0235)
IND_AV_DONATION_EXP	−.0461(.025)*		−.0466(.025)*	
COL_DONATION_EXP		.0202(.027)		.0153(.0271)
COL_FRUSTRATION		−.0248(.027)		−.0566(.031)*
SRA_0	−.404(.057)***	−.258(.057)***	−.401(0.057)***	−.251(.056)***
CONSTANT	1.200(.186)***	−.532(.196)***	1.226(.189)***	.713(.213)***
Adjusted R^2	.260	.158	.262	.177
F-stat.	23.27***	9.03***	17.62***	8.20***
N	203	197	203	197

*** 1% level of significance.
** 5% level of significance.
* 10% level of significance.

theory of expressive voting does not provide a consistent explanation for positive impact of proposing the full 10 € for donation on the subjects' short-run affective state.

4.2. Analyzing the behavioral data

Given these contradictory pieces of evidence, it seems reasonable to try to search for the distinct characteristics of those subjects who proposed to donate the full 10 €. For this purpose, we run multiple regressions. The endogenous variable is the probability of proposing 10 € for donation. The set of exogenous variables includes indicators for different motives for pro-social behavior stated in the literature and a number of control variables that have proven relevant in previous studies. Even though our major focus rests on treatment COL, we run these regressions to explain subjects' behavior in both treatments. The results on treatment IND serve as a benchmark for the results obtained in the regressions for treatment COL.

First, we generate two variables related to the Kinderschutzbund by asking subjects to state their level of agreement to the following statements: (1) "I consider the purpose of the Kinderschutzbund to be very important.", (2) "In Germany, child protection is guaranteed by the government to a sufficient degree." A 5-point scale is used from (5) "I agree fully" to (1) "I disagree entirely". The dummy variables IMPORT and SECURE take on the value 1 for subjects stating a value higher than 3, otherwise it is 0. We expect a positive influence for IMPORT and a negative influence for SECURE on the probability of proposing resp. donating the full 10 €.

We control for the fact that subjects' may consider it their moral duty of society to help children in need (e.g., Kotzebue & Wigger, 2010; Roemer, 2014). We account for the impact of moral convictions by introducing a variable that captures subjects' attitude toward Kant's Categorical Imperative. The German philosopher Immanuel Kant (1724–1804) is one the most influential philosophers in the realm of deontological ethics (e.g., Beauchamp, 2000). He proposed the well-known Categorical Imperative: An individual shall "[...] *act only on that maxim through which you can at the same time will that it should become a universal law*." (Kant, 1907: 88). We ask participants to what degree one should follow this guideline when making decisions (10 point scale). If a participant chose a value of 8 or higher, we classify him as Kantian. In this case, the dummy variable KANT takes on the value 1 (0 else). Given that helping children in need is widely accepted as a positive thing to do, we expect Kantian subjects to arrive at the conclusion that every subject should donate at least some of the 10 € to the KSB. In treatment COL, subjects can propose exactly this: Every subject should donate a certain amount *X* to the KSB. The individual subject can make this proposal at negligible material costs because he cannot expect to be pivotal. As the lack of pivotality holds for all subjects, however, it is not clear ex ante that Kantian subjects should propose a higher *X* for the collective donation than non-Kantian subjects. This is entirely different for treatment IND, because the Categorical Imperative strictly forbids free-riding (e.g., Roemer, 2014). If a subject arrives at the conclusion that every subject should donate a certain amount *X*, the Categorial Imperative demands to donate exactly this amount individually. Thus, in treatment IND, we expect Kantian subjects to be more likely to donate the full 10 € than non-Kantian subjects – other things equal.

We control for subjects' gender because it is found to play an important role in previous studies (e.g., Corneo & Grüner, 2002; Piper & Schnepf, 2008). We introduce a FEMALE-dummy that takes on the value 1 for female subjects and 0 for males. Females are expected to be more likely to propose resp. donate the full 10 €. In addition, we ask subjects whether they work during the semester. The dummy variable WORK takes on the value 1 for subjects who work for money during the semester and 0 for subjects who do not work. Subjects who work are expected to receive less support from their families and/or the government or have a higher preference for present private consumption. We expect a negative sign. A substantial share of subjects enter university after having successfully passed vocational training. The dummy variable JOB_EDU takes on the value 1 for these subjects (0 else). Subjects with job-education can expect higher hourly wages and – other things equal

Table 3
Probit regressions on behavioral data by treatment – baseline model.

End var.	IND_DONATED = 10	IND_DONATED = 10 (marg. eff.)	COL_PROPOSED = 10	COL_PROPOSED = 10 (marg. eff.)
IMPORT	.4284(.2617)	.1183(.0663)*	.1016(.2452)	.032(.0758)
SECURE	−.6418(.3279)*	−.1632(.06877)**	−.0452(.2974)	−.0143(.0933)
KANT	.5307(.2461)**	.1673(.0803)**	.3573(.2165)*	.1203(.0754)
FEMALE	.2346(.2015)	.0695(.0601)	−.0446(.21)	−.0143(.0673)
WORK	−.3174(.2292)	−.0904(.9627)	−.0273(.2058)	−.0087(.0656)
JOB_EDU	.646(.2135)***	.1896(.0596)***	.1793(.208)	.0581(.0678)
NON-GERMAN	.8183(.2968)***	.2633(.0956)***	−.0329(.3001)	−.0104(.0947)
NGO_T	.246(.2079)	.0727(.0618)	.1597(.1964)	.0514(.0634)
MEDIA_T	−.7305(.2855)**	−.1866(.06)***	−.2991(.2749)	−.09(.0767)
BERNBURG	.0212(.2288)	.0062(.0674)	.1889(.223)	.0619(.0744)
CLOSE			−.2733(.233)	−.0841(.0683)
CONSTANT	−1.4271(.3155)***		−.836(.2928)***	
Pseudo R^2	.1239		.0339	
χ^2	30.59***		8.02	
N	209		202	

*** 1% level of significance.
** 5% level of significance.
* 10% level of significance.

– have a higher income. In addition, they are older on average and collected life-time experiences beyond school and universities that the other subjects often lack. We differentiate between German and non-German subjects. The dummy NON_-GERMAN takes on the value 1 for non-Germans and 0 for Germans.[5] We also introduce subjects' trust in NGOs and in the media. It is measured by using a scale from 1 (very much trust) to 4 (no trust). If subjects stated to have very much (1) or much trust (2), the variable NGO_T resp. MEDIA_T, is 1; otherwise it is 0. We expect trust in NGOs to lead to a higher probability of proposing resp. donating the full 10 € but we have no predictions with respect to the expected sign of the MEDIA_T variable. In addition, we introduce the session-dummy BERNBURG to account for possible session-effects.[6]

In the questionnaire filled in after the experiment, we asked subjects "What share of subjects in your group do you expect to propose an amount to be donated collectively that exceeds the amount you proposed?" Based on the answer to this question (SHARE_HIGHER), we construct the dummy CLOSE. It takes on the value 1 for subjects for whom SHARE_HIGHER lies between 40% and 60% (0 else). CLOSE is included to control for a possible impact of the expectation of being pivotal. Including CLOSE is merely an act of caution because the expectation to be pivotal does not change the incentive to propose a high collective donation in our experimental set-up (see Section 3).

We run a probit model with the endogenous variable taking on the value 1 for subjects proposing the full 10 € (0 else) in treatment COL. The analogous regression is run for treatment IND. Here, the endogenous variable takes on the value 1 if subjects donate the full 10 € individual (0 else). The results from treatment IND serve as a benchmark for the results for treatment COL. The baseline regression model in Table 3 contains all variables described above for treatment IND and adds the variable CLOSE for treatment COL.

For treatment IND, we find a significantly positive impact of JOB_EDU, KANT, NON_GERMAN on the probability to donate the full 10 €. A significantly negative coefficient estimator is found for SECURE and MEDIA_T. These results are generally in line with the predictions developed above. Among the significant variables, NON_GERMAN has the largest marginal effect. At the same time, JOB_EDU, KANT, SECURE, and MEDIA_T have a marginal effect of more than 15 percentage points. We see a distinctly different picture for treatment COL: None of the explanatory variables yield significant coefficient estimators and the χ^2-statistic is insignificant (even at the 10% level). Appendix A contains a number of additional models with a larger set of variables and additional sensitivity analyses. The results stated here hold for the models presented in Appendix.

In sum, the regressions for treatment IND deliver a number of significant coefficients that are in line with the predictions. The overall explanatory value is significant. For treatment COL, however, none of the factors included contribute to explaining why some subjects propose to donate the full 10 € while others do not.[7]

[5] Most non-Germans have roots in Eastern Europe or Turkey. They are on average poorer than the Germans but there is no straightforward prediction regarding their behavior because of religious and ethnic differences between the two sub-groups. Unfortunately, splitting the sample further into these sub-groups left us with too few observations to draw reliable conclusions.

[6] We check for possible collinearity among explanatory variables and find the coefficients of direct correlation are all below 0.3. Correlation matrices are available with the authors upon request.

[7] This central result is supported by tobit-regressions that use the continuous variables IND_DONATED and COL_PROPOSED as endogenous variables and control for the lower limit of 0 € and the upper limit of 10 €. For two reasons, however, the tobit results are not reported here. First the relevant tests indicate that the essential assumptions underlying the probit approach are violated (see Cameron & Trivedi, 2009: 535–539). Second, the results in Section 4.1 suggest that essential line of distinction is the one between those subjects who propose to donate the full 10 € and those who propose less. The probit-approach reported above is the suitable empirical model to capture this distinction. The regression may be interpreted as the first step in a hurdle model. Engel (2011) uses hurdle models to show an interesting aspect of human behavior in dictator games. The decision whether or not to give a positive amount and the factors driving the decision precisely which positive amount to give are driven by different factors. As these questions are not the main focus of the paper and the overall performance of the first-step regressions for treatment COL is insignificant, we do not perform hurdle model regressions here.

5. Conclusion

In this paper, we present an economic experiment that compares the change in short-run affect resulting from donations to charity in two different settings. In the settings where subjects decide about the size of their donation individually, we find evidence for a warm-glow of giving the full endowment individually. In the setting where subjects decide collectively about how much to donate, we find no evidence that the expected collective donation causes a feeling of warm-glow. On the other hand, proposing to donate the full endowment of 10 € is found to yield a significant improvement in subjects affective state. Further analyses cast doubt on the notion that this pattern can be interpreted as evidence for expressive motives. In addition, they do not yield systematic relationship between subjects' personal characteristics, attitudes and moral preferences and their inclination to propose the full 10 € for donation. Summing up, we find evidence that proposing high collective donations is an individually rewarding act but we have little to say about why some subjects propose a high collective donations while others do not.

As a side-result, our paper contributes to the literature on the role of moral convictions in the individual decision to contribute to a common cause: Subjects' who consider Kant's Categorical Imperative to be an important guideline for individuals decisions are more likely to abstain from free-riding and donate the full 10 € to the KSB. This result holds after subjects' views on the importance KSB's work – a proxy for the existence of other-regarding preferences – has been controlled for. It supports the view of Roemer (2014). Accordingly, subjects seem to apply Kantian thinking when analyzing the consequences of their own action and this Kantian thinking is an independent factor that reduces free riding in collective dilemmas –regardless of whether subjects entertain other-regarding preferences.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.joep.2015.09.001.

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